

RYAN PEACH

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CONTACT

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PROFILE

OBJECTIVE: Seeking to move south and continue a career in Computer Science and Electrical Engineering.

AVAILABILITY: Beginning Summer 2016

EDUCATION & AWARDS

Masters of Science in Computer Science, Spec. Interactive Intelligence, Robotics, & Perception December 2017

Georgia Tech. Institute of Technology, Atlanta, GA

Bachelor of Science in Electrical Engineering, Mathematics Minor December 2014

Western Kentucky University, Bowling Green, KY - GPA: 3.4

WORK EXPERIENCE

Nasco-NHK, Bowling Green, KY May 2015 -

Electrical Engineer (Controls & Automation) Present

- ◆ Programmed PLC's and robotics for a variety of control & automation applications.
- ◆ Designed cabinets using AutoCAD, implemented hardware upgrades to equipment.
- ◆ Diagnosed and maintained industrial electrical equipment.

Institute for Combustion Science and Technology, WKU

Electrical Engineer (Signals & Amplification)

January 2015 -
May 2015

- ◆ Worked with physicists and other researchers in designing rigs for experimentation.
- ◆ Helped design applications of Atomic Force Microscopes, Piezoelectric material, Signal Amplifiers, and Terra-hertz Lasers. Designed PCB's & programmed FPGA's.

Applied Physics Institute, Center for Research & Development, WKU

Student Electrical Engineer (PCB Design, Embedded Systems)

May 2014 -
December 2014

- ◆ Designed PCB's, performed research for both scientific and consumer purposes.
- ◆ Implemented chemical sensor arrays and machine learning algorithms.

Department of Chemistry, Western Kentucky University

Student Software Developer / Electrical Engineer

January -
March, 2014

- ◆ Designed software to control local multimeter via user friendly LabVIEW computer interface.
- ◆ Created applications for graphing Voltage, Resistance, & Current Density in real-time.
- ◆ Designed testing rig and product enclosure for solar array.

DHS-STEM Student Internship, Pacific Northwest National Laboratory

Summer Intern (Social Data Analyst)

June -
August, 2013

- ◆ Ran discrete-time Fourier analyses, digital filtering algorithms, and statistical analyses on large time-series datasets in Python and MatLab.
- ◆ Implemented machine learning spectral algorithms on large NLP datasets.

Lego Robotics Camp, Center for Gifted Studies, WKU

Robotics Teacher (Lego Mindstorms, NXT Programming)

June, 2012;
June, 2014

KEY SKILLS

Java, Scala, Python (Numpy, SciPy, Pandas, etc.), C/C++, Assembly, PLC Ladder Logic, Structured Ladder, Structured Text, SFC, LabVIEW (CLAD), MatLab, Simulink, Git, LaTeX, Linux, Android, HTML/CSS, VHDL, Verilog, OpenSCAD, 3D Printers

RS232/485, Modbus, Arduino, Allen-Bradley, Mitsubishi, Motoman, Proface HMI's, AutoCAD, MultiSIM, Eagle, Maxwell 3D, Unix, Linux, Windows, MS Office, Electronics / PCB Design, Microprocessors, AC/DC Motor Control

Robotics, Computer Vision, Artificial Intelligence, Machine Learning, ASIC FPGA Design, Physics & Mathematics, Computer Programming, Control Systems, Signal Analysis, High Quality Signal Amplification, Digital Signal Acquisition, Networking

SCHOOL & PROFESSIONAL PROJECTS

SCADA Senior Design Project (WKU)

Fall 2013 -
Spring 2014

- ◆ Created three PLC testbed systems: a traffic light, a water pump, and a power grid.
- ◆ Remotely controlled each system via SCADA.
- ◆ Performed simulated attacks on each system, and creating contingency programming to both prevent and react to each attack.

Motor Drives Control Project (WKU)

Spring 2013

- ◆ Contributed to the writing of nine LabVIEW scripts to control and measure the speed, power, torque, voltage, and current supplied to/measured by a motor drive system.
- ◆ Created host-controlled VIs for top-down processing, and user interfaces allowing the user to play with each of the variables described above either independently (open loop), or with computer assistance (closed-loop).
- ◆ Created a LabVIEW library for the open-loop and closed-loop control of DC and AC motors, along with DC Generators, for easy programming of future VIs on the project.

Maxwell 3D Simulation (WKU)

Fall 2012

- ◆ Collaborated on a Maxwell 3D project to create an animated simulation of the magnetic field lines around an ideal 3-phase induction motor under various loads.
- ◆ Both no load and full load simulations generated.

Digital Clock Design (WKU)

Fall 2011

- ◆ Design a 2 digit stopwatch style digital clock using Eagle and MultiSIM.
- ◆ Print the circuit board using the circuit printing facility and build the board.
- ◆ Extra Credit: Design a 4 digit minute/seconds stopwatch instead.

Robot Competitions (WKU, Georgetown College)

Fall 2010

- ◆ Taught Mindstorm robotics for over seven years implementing advanced programming challenges such as maze solving algorithms, as well as teaching basic mechanical and electrical concepts.
- ◆ Designed a robot using a Basic Stamp and Light Sensors capable of quickly following a line through an obstacle course. Created the body of the robot using machining facility.
- ◆ First Place in Freshman speed competition.

PERSONAL PROJECTS

3DBook - 3D Printed Braille Literature

Winter 2015

- ◆ Open-source software written in OpenSCAD and Python used for the generation of 3D Printer (.stl) files containing full text documents written in braille, printed and bound.
- ◆ Enables blind individuals worldwide to access any literature with readily available tools.
- ◆ Suggested for admission into the Touch of Genius award by National Braille Press.

OpenScan - Mobile Document Scanner

Winter 2015

- ◆ Developed collaborative international open-source C++ and Android Application.
- ◆ Scans pages and QR codes via video / camera interface quickly and accurately using a variety of methods. Capable of scanning numerous documents with ease in a single use.
- ◆ Gives user choice over all settings and access to debugging views. Soon to implement AI to detect proper values for ease of use.

Symbolic Circuit Simulator

Spring 2012

- ◆ Uses Python and Sympy to perform s-Domain circuit analysis
- ◆ Useful for deriving the laplacian transformation between input and output of filters and transformer circuits symbolically.

Planetary Orbit Simulation

Spring 2013

- ◆ Uses MatLab and Newton's law of universal gravitation to plot the motion of any number of gravitationally effected bodies.
- ◆ Uses parallel processing and MatLab commands to calculate, export, and plot position data.

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